

Resistor Colour Codes

				
Gold	—	—	÷10	5% tolerance
Silver	—	—	÷100	10% tolerance
Black	0	0		
Brown	1	1	0	1% tolerance
Red	2	2	00	
Orange	3	3	000	
Yellow	4	4	0000	
Green	5	5	00000	
Blue	6	6	000000	
Violet	7	7	0000000	
Grey	8	8		
White	9	9		

EXAMPLES	
	1K (1,000Ω) 5%
	4R7 (4.7Ω) 5%
	1M8 (1,800,000Ω) 5%
	220K (220,000Ω) 5%

Resistor Values for LEDs assuming approximately 20mA is required Suggested values for resistors when using LEDs with batteries or power supplies	
Voltage	Value
3V	120Ω
5V	220Ω
9V	470Ω
12V	560Ω

For a four band resistor:

- the first band denotes the first digit of the resistance value
- the second band denotes the second digit of the resistance value
- the third band represents the multiplier
- the fourth band signifies tolerance

The resistor shown above with brown, black, red and gold bands has resistance value of $1000\Omega \pm 5\%$ (brown black red = 102 = 10×10^2). The gold band denotes tolerance of 5%. It means, its resistance value can range between 1050Ω ($= 1000\Omega + 5\%$ of 1000Ω) to 950Ω ($1000\Omega - 5\%$ of 1000Ω).